ECE 3441 Homework **Unit 14** **Name/ID#**______________________

14.4 – 14.9, 14.15, 14.19, 14.22, 14.33, 14.47

14.4 A sequential circuit has one input and one output. The output becomes 1 and remains 1 thereafter when at least two 0's *and* at least two 1's have occurred as inputs, regardless of the order of occurrence. Draw a state graph (Moore type) for the circuit (nine states are sufficient). Your final state graph should be neatly drawn with no crossed lines.

14.5 A sequential circuit has one input (X) and two outputs (Z_1 and Z_2). An output $Z_1 = 1$ occurs every time the input sequence 010 is completed, provided that the sequence 100 has never occurred. An output $Z_2 = 1$ occurs every time the input 100 is completed. Note that once a $Z_2 = 1$ output has occurred, $Z_1 = 1$ can never occur but *not* vice versa. Find a Mealy state graph and state table (minimum number of states is eight).

14.6 A sequential circuit has two inputs (X_1 and X_2) and one output (Z) . The output begins as 0 and remains a constant value unless one of the following input sequences occurs:

a. The input sequence $X_1X_2 = 01, 00$ causes the output to become 0 .

b. The input sequence $X_1X_2 = 11,\ 00$ causes the output to become 1 .

c. The input sequence $X_1X_2 =\ 10, 00$ causes the output to change value.

Derive a Moore state table.

14.7 A sequential circuit has one input (X) and one output (Z) .

Draw a Mealy state graph for each of the following cases:

a. The output is $Z = 1$ iff the total number of 1's received is divisible by 3 .
(*Note*: $0, 3, 6, 9, \ldots$ are divisible by 3 .)

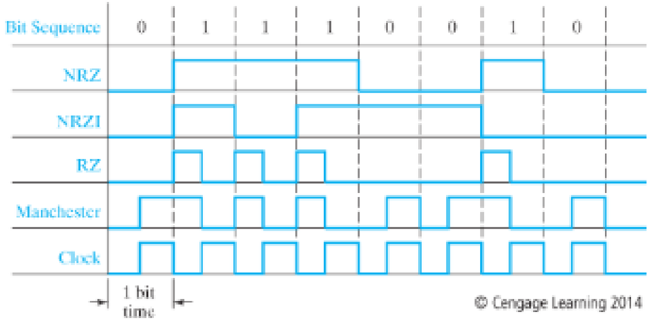
b. The output is $Z = 1$ iff the total number of 1's received is divisible by 3 *and* the total number of 0's received is an even number greater than zero (nine states are sufficient).

14.8 A sequential circuit has two inputs and two outputs. The inputs (X_1 and X_2) represent a 2-bit binary number, N . If the present value of N is greater than the previous value, then Z_1 is 1 . If the present value of N is less than the previous value, then Z_2 is 1 . Otherwise, Z_1 and Z_2 are 0 . When the first pair of inputs is received, there is no previous value of N , so we cannot determine whether the present N is greater than or less than the previous value; therefore, the "otherwise" category applies.

a. Find a Mealy state table or graph for the circuit (minimum number of states, including starting state, is five).

b. Find a Moore state table for the circuit (minimum number of states is 11).

14.9 **(a)** Derive the state graph and table for a Mealy sequential circuit which converts a serial stream of bits from NRZ code to NRZI code. Assume that the clock period is the same as the bit time as in Figure 14-20.



(b) Repeat (a) for a Moore sequential circuit.

(c) Draw a timing diagram for your answer to (a), using the NRZ waveform in Figure 14-20 as the input waveform to your circuit. If the input changes occur slightly after the clock edge, indicate places in the output waveform where glitches (false outputs) can occur.

(d) Draw the timing diagram for your answer to (b), using the same input waveform as in (c).

14.15 a. A Mealy sequential circuit has one input (x) and one output (z). z can be 1 when the fourth, eighth, twelfth, etc. inputs are present, and $z = 1$ if and only if the most recent input combined with the preceding three inputs was not a valid BCD encoding of a decimal digit; otherwise, $z = 0$. Assume the BCD digits are received *least* significant bit first. Derive a state table for the circuit. (Six states are sufficient.)

b. Repeat for a Moore circuit (i.e., $z = 1$ if and only if, after the fourth, eighth, twelfth, etc. inputs have been received, the previous four inputs were not a valid BCD digit).

c. Is it possible for a Moore circuit to generate the correct output while the fourth input bit is present rather than after it has been received? Explain your answer.

14.19 **a.** A Mealy sequential circuit has one input (x) and one output (z). z can be 1 when the fourth, eighth, twelfth, etc. inputs are present, and $z = 1$ if and only if the most recent input, combined with the preceding three inputs, was not a valid excess- 3 encoding of a decimal digit; otherwise, $z = 0$. Assume the excess- 3 digits are received *least* significant bit first. Derive a state table for the circuit. (Nine states are sufficient.)

b. Repeat for a Moore circuit (i.e., $z = 1$ if and only if, after the fourth, eighth, twelfth, etc. inputs have been received, the previous four inputs were not a valid excess- 3 digit). (Ten states are sufficient.)

c. Is it possible for a Moore circuit to generate the correct output while the fourth input bit is present rather than after it has been received? Explain your answer.

14.22 A sequential circuit has one input (X) and two outputs (Z_1 and Z_2). An output $Z_1 = 1$ occurs every time the input sequence 100 is completed provided that the sequence 011 has never occurred. An output $Z_2 = 1$ occurs every time the input 011 is completed. Note that once a $Z_2 = 1$ output has occurred, $Z_1 = 1$ can never occur but *not* vice versa. Find a Mealy state graph and state table (minimum number of states is eight).

14.33 A Moore sequential circuit has one input and one output. The output is 1 if and only if both of the following conditions are met:

a. The input sequence contains exactly two groups of 1's , and

b. Each of these groups contains exactly two 1's .

Each group of 1's must be separated by at least one 0 . A single 1 is considered a group of 1's containing one 1 . For example, the sequence

$$X = \;\; 0 \;\; 1 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 1 \;\; 0 \;\; 1 \;\; 1 \;\; 1 \;\; 0$$

satisfies both conditions after the first two pairs of 1's . However, when more 1's appear, condition (a) is no longer satisfied. Therefore, the output sequence should be

$$Z = \;\; (0) \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 0$$

On the other hand, the sequence

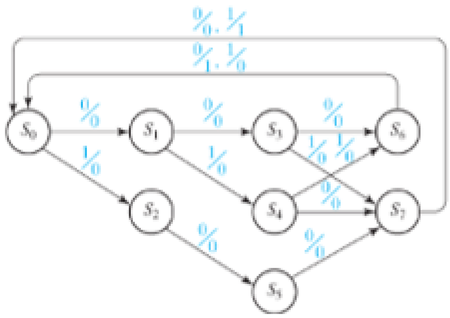
$$X = \;\; 1 \;\; 0 \;\; 1 \;\; 1 \;\; 0 \;\; 1 \;\; 1 \;\; 0$$

never satisfies condition (b), because the first group of 1's contains only one 1 . Besides, after the second pair of 1's , (a) is no longer satisfied because the input sequence contains three groups of 1's . Therefore, the output should always be 0 .

$$Z = \;\; (0) \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0$$

Derive a state graph and table.

14.47 Modify the BCD parity detector of Table 14-9 for the case where the least significant bit of the BCD digits is received first. Construct the incompletely specified Mealy model state table. (*Hint*: It can be constructed by specifying states to remember the first 3 bits of a BCD digit, but not all sequences of length three have to be distinguished. It requires at most 11 states.)



	Next State		Output	
Present State	$X=0$	$X=1$	$X=0$	$X=1$
S_0	S_1	S_2	0	0
S_1	S_3	S_4	0	0
S_2	S_5	-	0	-
S_3	S_6	S_7	0	0
S_4	S_7	S_6	0	0
S_5	S_7	-	0	-
S_6	S_0	S_0	1	0
S_7	S_0	S_0	0	1